

AKNUR KALILAYEVA

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Education

University of Illinois at Urbana-Champaign

Aug 2025 - May 2027

Master of Science in Statistics (Analytics), GPA: 3.7/4.0

Coursework: Statistical Modeling, Statistical Learning, Regression Analysis, Applied Bayesian Analysis, Time Series Analysis

Nazarbayev University

Jun 2024

Bachelor of Science in Mathematics (Magna Cum Laude Honors), GPA: 3.80/4.00, Top 2/93

Coursework: Statistical Programming, Image Processing, Machine Learning, Linear Programming, Mathematical Statistics, Optimization, Probability, Actuarial Science 1&2 (Exam P, Exam FM), Microeconomics, Macroeconomics, Stochastics

Work Experience

Lead Technology Assistant – ATLAS

Aug 2025 - present

- **Reduced first-response KPI 30%** by standardizing ticket triage workflow and automating response playbooks using templates/macros, reducing handling time and inconsistency.
- **Improved user satisfaction by 27%** by building documentation for recurring issues and publishing quick-reference troubleshooting guides to improve self-serve resolution and repeatability.
- Resolved **5-10** faculty/student support tickets per day across classroom technology, software access, and accounts.

Research Assistant — Department of Mathematics

Aug 2022 — May 2025

- Built a quarterly credit-risk analytics pipeline in **Python**: resampled portfolio default rates to quarterly frequency, aligned with **55+ macroeconomic indicators**, and performed data cleaning, EDA, and feature screening.
- Trained and evaluated regression models (scaling, correlation-based feature selection, **IQR outlier filtering**) to forecast default-rate movements; achieved **RMSE 0.020–0.023** on holdout.

Data Analyst — Data Lab

Jun 2023 — Jul 2023

- Developed ML risk-segmentation models (**Logistic Regression, Gradient Boosting**) with feature selection (**RFE/variance filtering**) and class-imbalance handling (**balanced class weights, SMOTE**).
- Improved discrimination vs baseline (**ROC AUC 0.714→0.744, PR-AUC 0.141→0.195**) and improved probability reliability via **isotonic calibration**; communicated results with confusion matrices and model reports.
- Built the modeling dataset using **SQL** joins/CTEs and automated extracts into Python for training and evaluation.

Head of PR Department, Marketing Manager – SIAM

Jan 2020 – May 2022

- **Organized and promoted 50+ mathematics-related events** across social media channels; crafted content calendars, visuals, increased sign-ups by **65%** using engagement analytics.
- Created 35 consulting-style presentation materials in **PowerPoint** and **Canva**, using charts, infographics, and clear visual narratives for event promotion, partner outreach, and internal reporting.

Skills

Programming Languages: Python, R, SQL, Matlab

Python/R Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scipy, PyTorch, TensorFlow, Scikit-learn, Statsmodels

Data Science Technologies: Data Cleaning, Data Mining, Forecasting, Regression, Classification, Experimental Design, A/B tests, Data Visualization, Dashboarding, Feature Engineering, Data quality checks, Pipelines/Automation

Data Processing: ETL, Snowflake, MS Excel, Power BI, Tableau

Development Tools: Git, Jupyter Notebooks, Anaconda

Publications

1. **Kalilayeva, A.**, Zhumashev, D., Wei, D., Perveen, A., & Talamona, D. (2024). Investigation of Stereolithography Additively Manufactured Components for Deviations in Dimensional and Geometrical Features. *Polymers*, 16(23).
2. **Kalilayeva, A.** Nonlinear Implicit Regression Fit of the Mortality Data by the Three Parameter Generalized Logistic Model. (2024). Capstone project. Nazarbayev University. Supervisors: Prof. Dongming Wei, Prof. Kerem Ugurlu.

Selected Projects

- **Business Survival Prediction (Yelp Restaurants)**, [GitHub](#)
 - Engineered structured + behavioral features from the Yelp Dataset (early popularity, chain status, local competition, ratings/review volume) and defined **business lifetime (months)** as the target.
 - Combined **unsupervised structure discovery** (PCA/K-means) with linear, nonlinear models; observed **test AUC 0.74 with gradient boosting vs 0.68 logistic/splines**, analyzed competition effects via LOESS, partial dependence.
- **SVD-based MNIST Digit Classification**, [GitHub](#)
 - Implemented an **SVD-based classification pipeline** for handwritten digits using the MNIST dataset.
 - Used low-rank representations to reduce dimensionality and support downstream classification experiments.

Leadership Experience

Teaching Assistant for **Calculus & Probability**

Study of the U.S. Institutes (SUSI), – Washington, USA

SUSI Global Leadership Workshop – Prague, Czechia

Merit-based Awards: Presidential Scholarship (top 1%), Skywalk of Fame (top 1%), Konrad Adenauer Foundation Sur-Place Scholarship (top 5%), Yessenov Scholarship (top 10%); Alumni Relief, Mereke Scholarships.